

KLOW — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	23mg, 80mg	9/13/2026 · Lot IWR-090926KL
4–15	1mg, 10mg, 21mg, 27mg, 42mg, 50mg, 73mg, 79mg, 80mg, 93mg	8/24/2026 · Lot IWR-07326079KL
16–23	1mg, 10mg, 12mg, 50mg, 152mg	4/15/2026



Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com |
summitbioanalytics.com

TESTED FOR

Iron Within Research

SAMPLE

KLOW — 80mg

PASS

BATCH #

IWR-
090926KL

PURITY

99.13%

MASS

81.23 mg

IDENTITY

Confirmed

COA #:

SBA-26-0179

Method:

Full QC Panel

Lot Number:

IWR-090926KL

Analysis Date:

2026-09-13

Labeled Content:

80mg

Date Received:

2026-09-07

Appearance:

White lyophilized
powder, intact seal

Vial Cap:

Clear #A4A2A7



SCAN TO VERIFY



Identity



Purity



Endotoxins



Fentanyl Free

Test	Specification	Result
Identity	KLOW	Confirmed
Purity (HPLC)	>= 98.0%	99.13%
Content (per vial)	Report only	81.23 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.04 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganus

Kasey Malganus

Lab Director

2026-09-13

COA #:

SBA-26-0179

Access Code:

79MHG8ZY

Analyst:

R. Jeffereson

Verify:

summitbioanalytics.com/verify

Issued:

2026-09-13

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Lot Number: **IWR-7626298-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **06/29/2026**
 Analysis Conducted: **06/24/2026**
 Searchable via: **horizonanalytical.com**

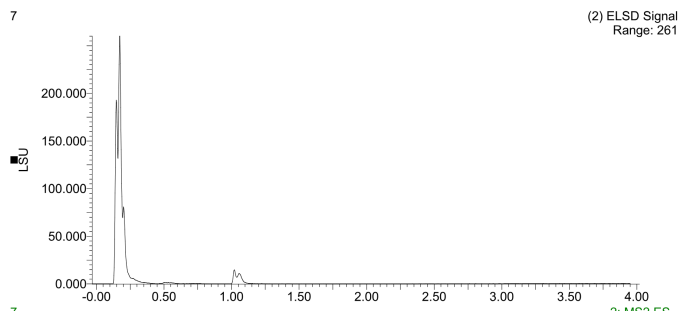
Compound:	BPC-157
Lot:	IWR-7626298-P
Appearance:	Blue Lyophilized Powder

CAS:	137525-51-0
Formula:	C ₆₂ H ₉₈ N ₁₆ O ₂₂
Mol Weight:	~1419.5 g/mol

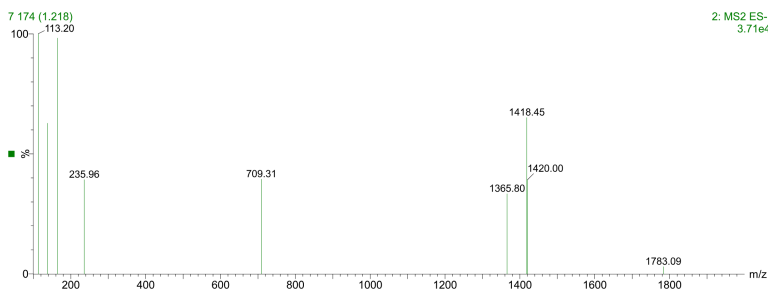
Pubchem CID: 108101

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	BPC-157	BPC-157	
Quantity:	10mg	9.93mg	
Purity:	≥ 98%	99.32%	



UPLC



MS

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

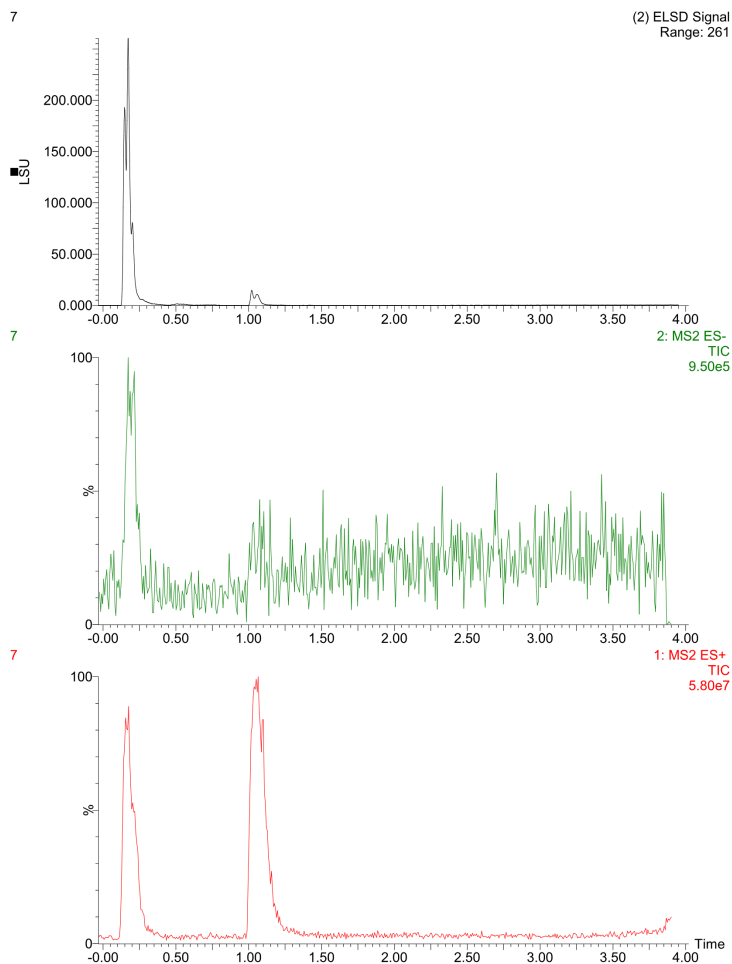


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

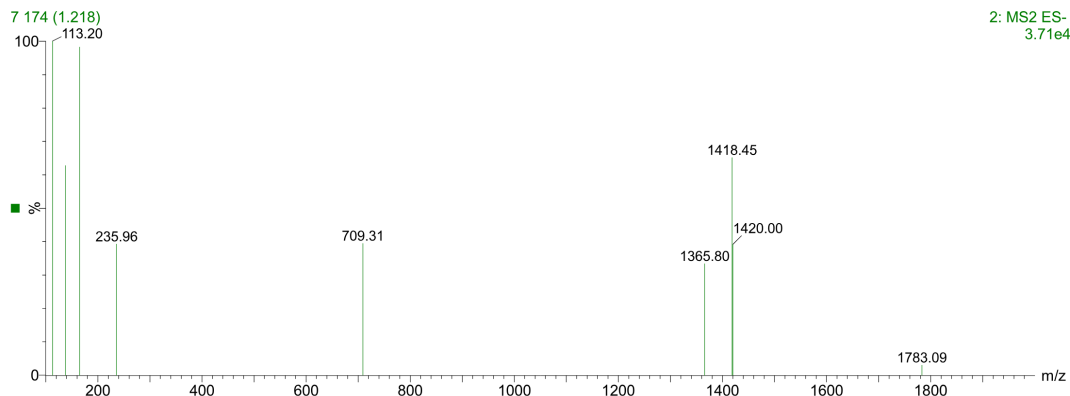
Lot Number: IWR-7626298-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
 Analysis Conducted: 06/24/2026
 Searchable via: horizonanalytical.com

BPC-157 (10mg) • Pubchem CID: 108101
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: **IWR-7626298-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **06/29/2026**
 Analysis Conducted: **06/24/2026**
 Searchable via: **horizonanalytical.com**

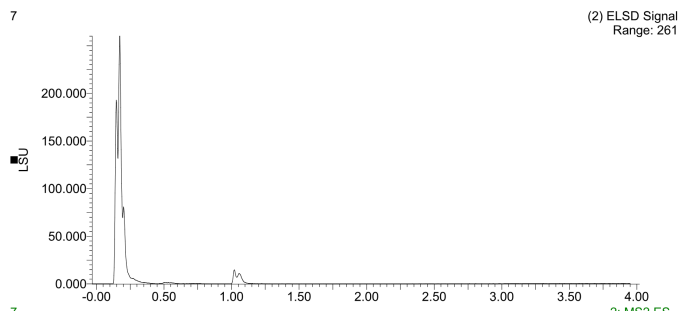
Compound:	TB-500
Lot:	IWR-7626298-P
Appearance:	Blue Lyophilized Powder

CAS:	77591-33-4
Formula:	C ₂₁₂ H ₃₅₀ N ₅₆ O ₇₈ S
Mol Weight:	~4963 g/mol

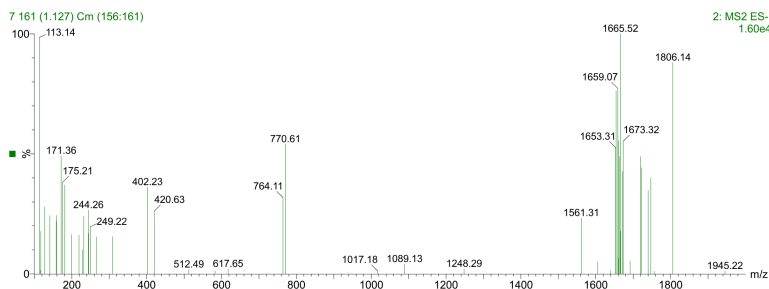
Pubchem CID: 16132341

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	TB-500	TB-500	
Quantity:	10mg	9.93mg	
Purity:	≥ 98%	99.2%	



UPLC



MS

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

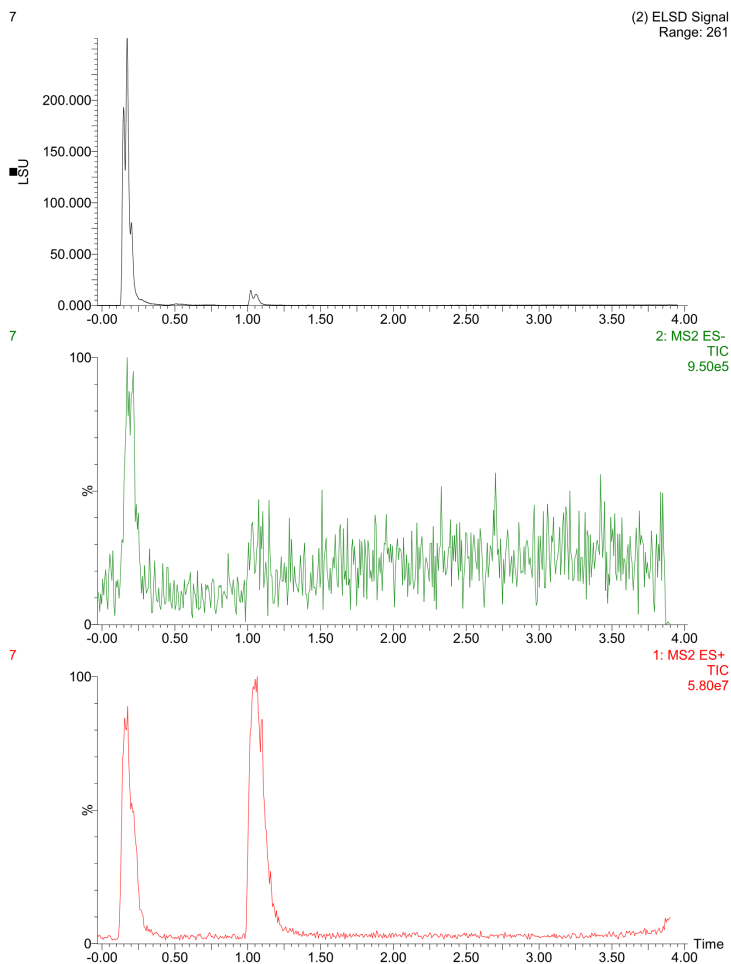


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

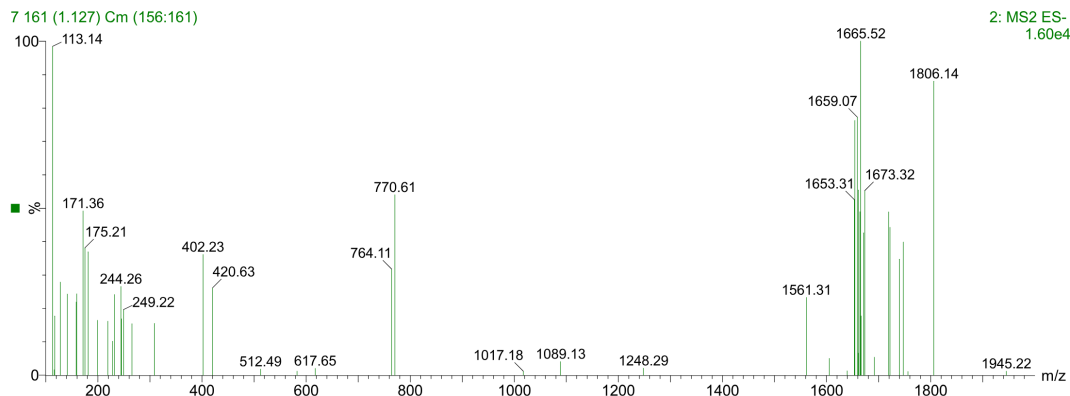
Lot Number: IWR-7626298-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
 Analysis Conducted: 06/24/2026
 Searchable via: horizonanalytical.com

TB-500 (10mg) • Pubchem CID: 16132341
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: **IWR-7626298-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **06/29/2026**
 Analysis Conducted: **06/24/2026**
 Searchable via: **horizonanalytical.com**

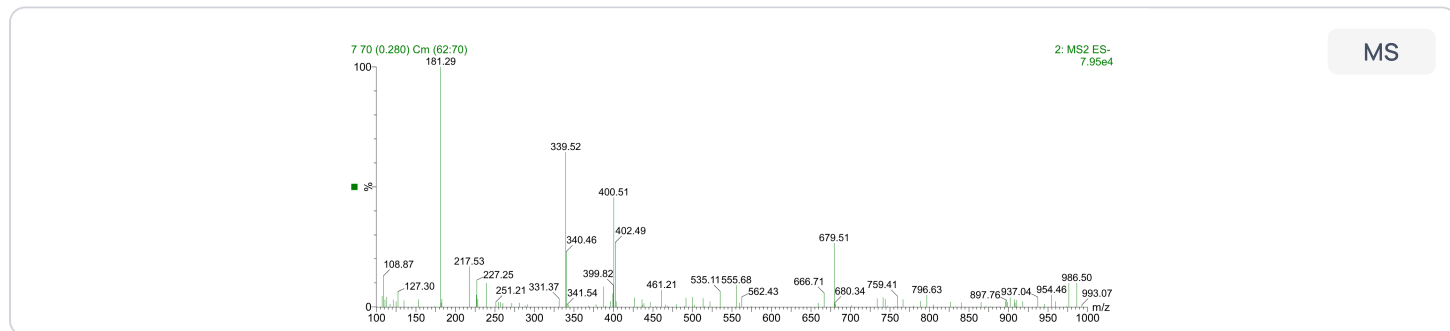
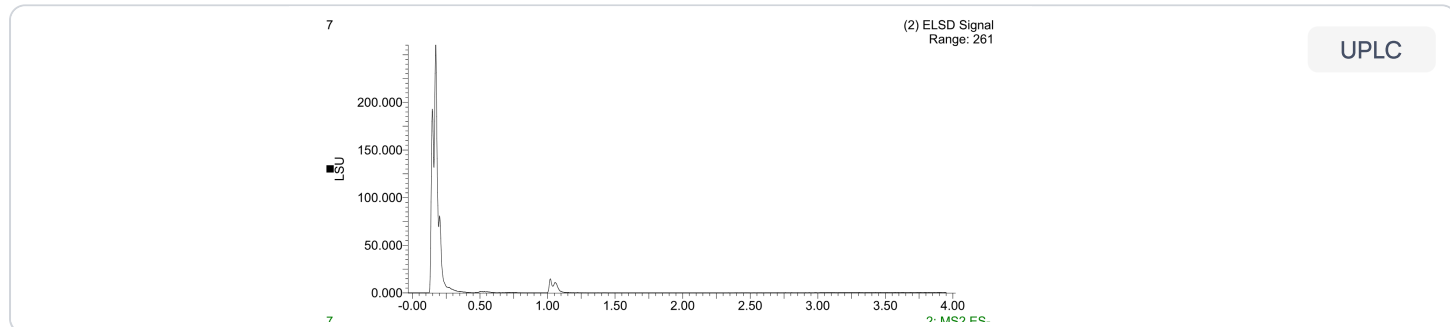
Compound:	GHK-Cu
Lot:	IWR-7626298-P
Appearance:	Blue Lyophilized Powder

CAS:	89030-95-5
Formula:	C ₁₄ H ₂₃ CuN ₆ O ₄
Mol Weight:	~402.92 g/mol

Pubchem CID: 71587328

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	GHK-Cu	GHK-Cu	
Quantity:	50mg	49.1mg	
Purity:	≥ 98%	99.51%	



Aleksey Yevtodiyyenko PhD
Research and Formulation Chemist

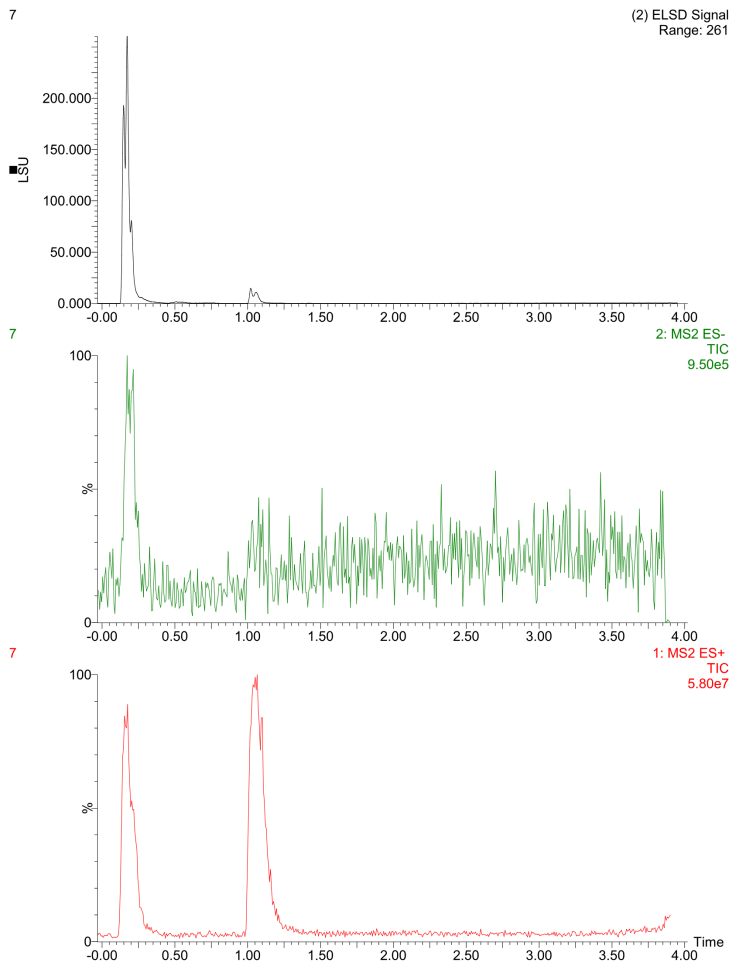


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

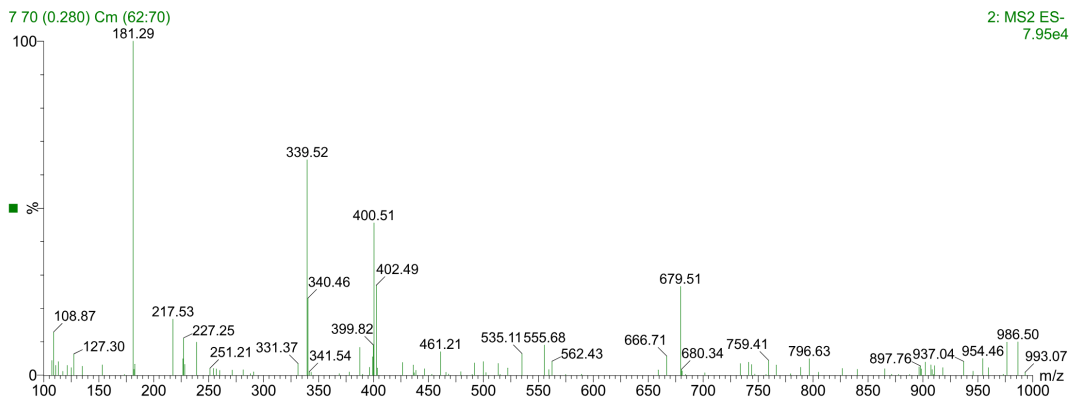
Lot Number: IWR-7626298-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
 Analysis Conducted: 06/24/2026
 Searchable via: horizonanalytical.com

GHK-Cu (50mg) • Pubchem CID: 71587328
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: **IWR-7626298-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **06/29/2026**
 Analysis Conducted: **06/24/2026**
 Searchable via: **horizonanalytical.com**

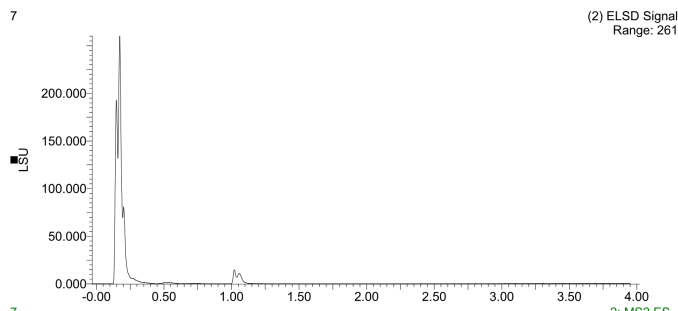
Compound:	KPV
Lot:	IWR-7626298-P
Appearance:	Blue Lyophilized Powder

CAS:	112965-21-6
Formula:	C ₁₇ H ₃₂ N ₆ O ₄
Mol Weight:	~384.48 g/mol

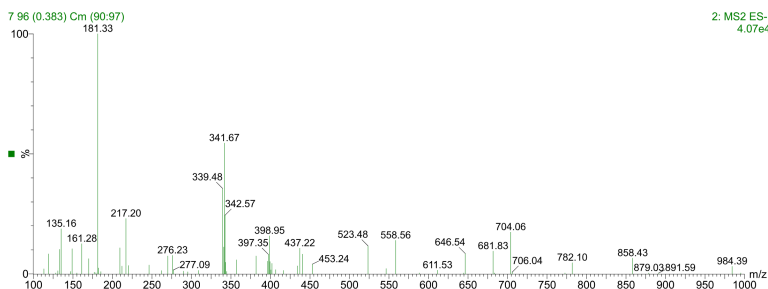
Pubchem CID: 9929972

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	KPV	KPV	
Quantity:	10mg	9.93mg	
Purity:	≥ 98%	99.32%	



UPLC



MS

Aleksey Yevtodiyyenko PhD
Research and Formulation Chemist

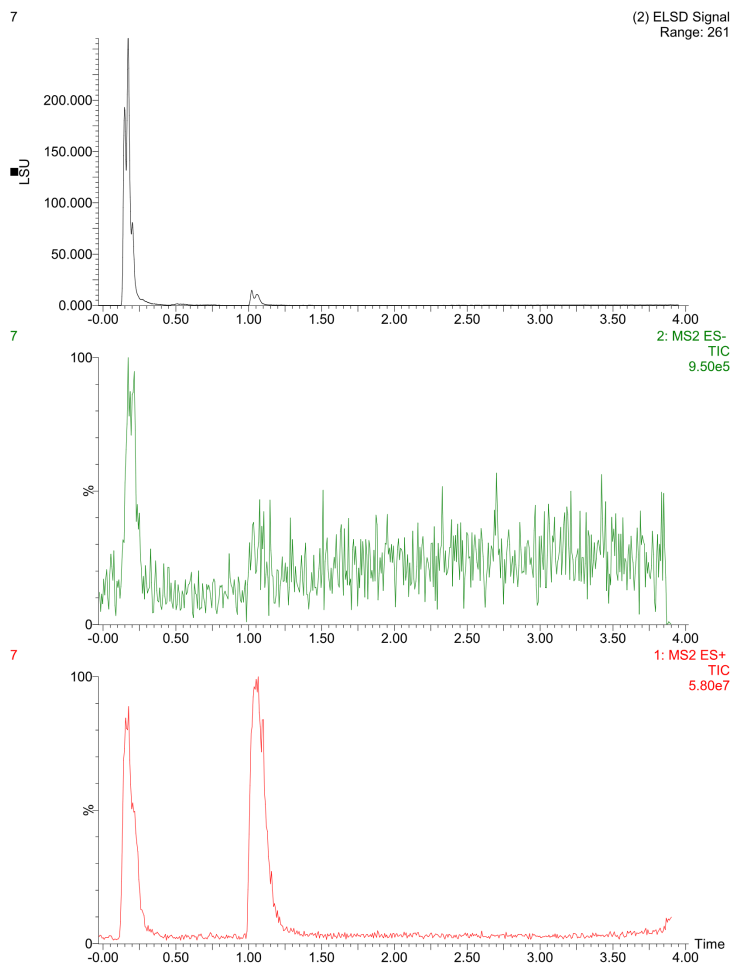


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

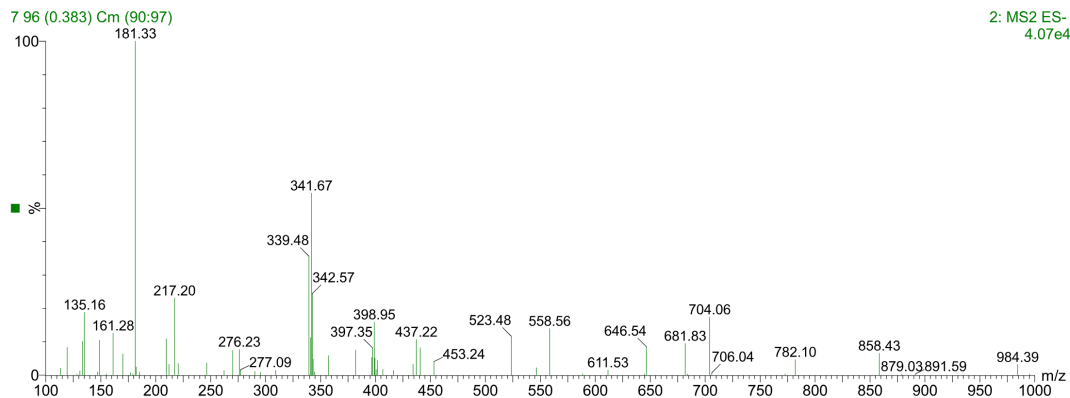
Lot Number: IWR-7626298-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
 Analysis Conducted: 06/24/2026
 Searchable via: horizonanalytical.com

KPV (10mg) • Pubchem CID: 9929972
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: [IWR-7626298-E](#)
Client Name: [Iron Within Research](#)
Identity: www.ironwithinresearch.com

Received Date: [06/29/2026](#)
Analysis Conducted: [06/24/2026](#)
Searchable via: horizonanalytical.com

Compound:	KLOW
Lot:	IWR-7626298-E
Appearance:	-

CAS:	BPC-157, TB-500, GHK-Cu, KPV
Formula:	N/A
Mol Weight:	N/A

Pubchem CID: N/A

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	KLOW	-	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist



This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0017

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com |
summitbioanalytics.com

TESTED FOR

Iron Within Research

SAMPLE

GHK-Cu / KPV / TB-500 / BPC-157 — 80 mg

PASS

BATCH #

IWR-2607-KLOW

PURITY

99.96%

MASS

78.21 mg

IDENTITY

Confirmed

COA #:

SBA-26-0017

Method:

Full QC Panel

Lot Number:

IWR-2607-KLOW

Analysis Date:

2026-08-02

Labeled Content:

80 mg

Date Received:

2026-07-29

Appearance:

**Lavender lyophilized
cake, intact seal**



SCAN TO VERIFY



Identity



Purity



Endotoxins



Fentanyl Free



VIAL CAP

Clear

#A4A2A7

CLOSURE AS
RECEIVED

Test	Specification	Result
Identity	GHK-Cu + KPV + TB-500 + BPC-157	All confirmed
Purity (HPLC, aggregate)	>= 98.0%	99.96%
Blend Composition	Report only	Quantified – see composition
Content (gross)	Report only	78.21 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.10 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

BLEND COMPOSITION Mass basis, per-component reference standards

COMPONENT	AMOUNT	SHARE OF TOTAL	IDENTITY
GHK-Cu	46.79 mg	59.83%	Confirmed
KPV	10.27 mg	13.13%	Confirmed
TB-500	10.73 mg	13.72%	Confirmed
BPC-157	10.42 mg	13.32%	Confirmed

Each component is quantified on a mass basis against its own reference standard. Share of total will not match a component's area-percent in the chromatogram — different chromophores respond differently at the monitoring wavelength. Amounts are reported as measured; this certificate does not assess them against a declared formulation.



Kasey Malganus
Lab Director
2026-08-02

COA #:

SBA-26-0017

Access Code:

RPFNDDA

Analyst:

R. Whitfield

Verify:

summitbioanalytics.com/verify

Issued:

2026-08-02

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Lot Number: **IWR-7964352-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **04/21/2026**
 Analysis Conducted: **04/16/2026**
 Searchable via: **horizonanalytical.com**

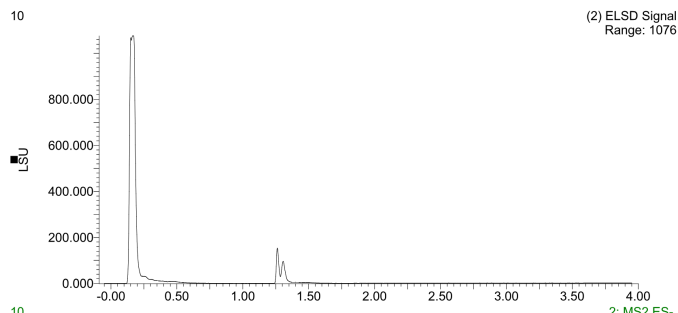
Compound:	BPC-157
Lot:	IWR-7964352-P
Appearance:	White Lyophilized Powder

CAS:	137525-51-0
Formula:	C ₆₂ H ₉₈ N ₁₆ O ₂₂
Mol Weight:	~1419.5 g/mol

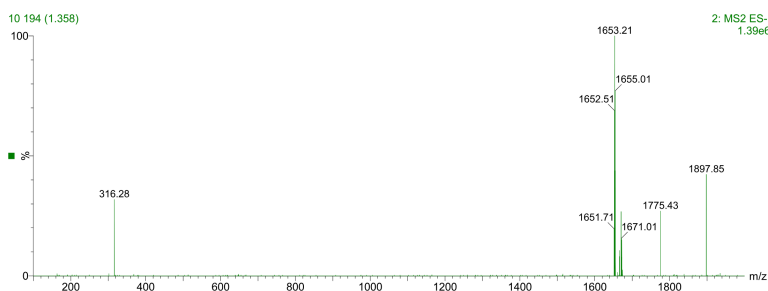
Pubchem CID: 108101

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	BPC-157	BPC-157	
Quantity:	10mg	9.75mg	
Purity:	≥ 98%	99.18%	



UPLC



MS

Aleksey Yevtodiyenko PhD
 Research and Formulation Chemist

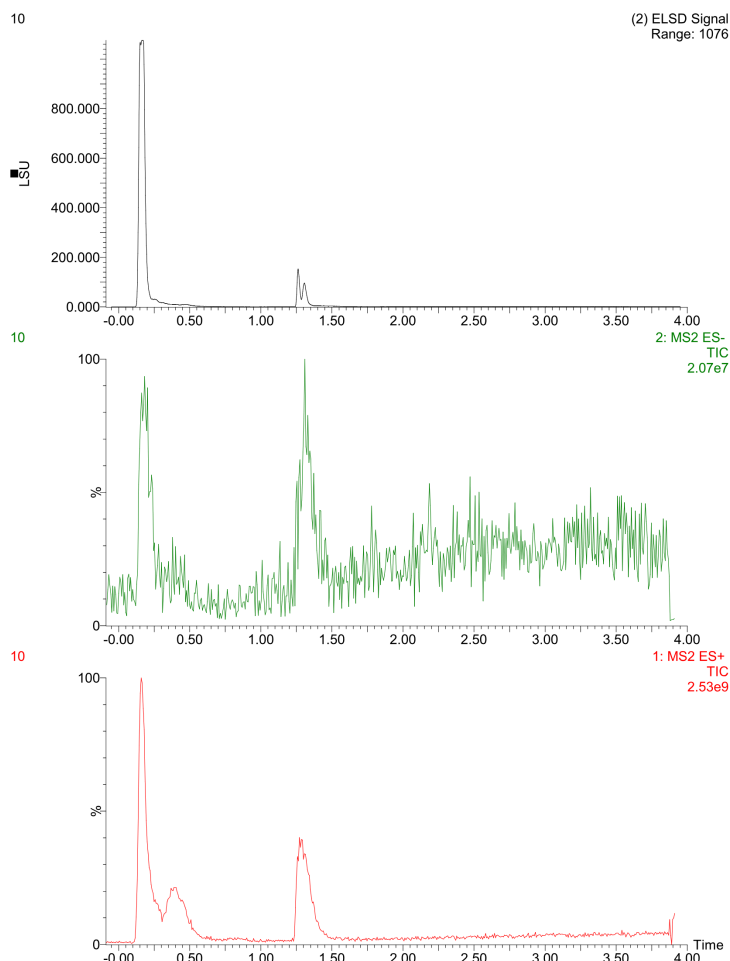


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

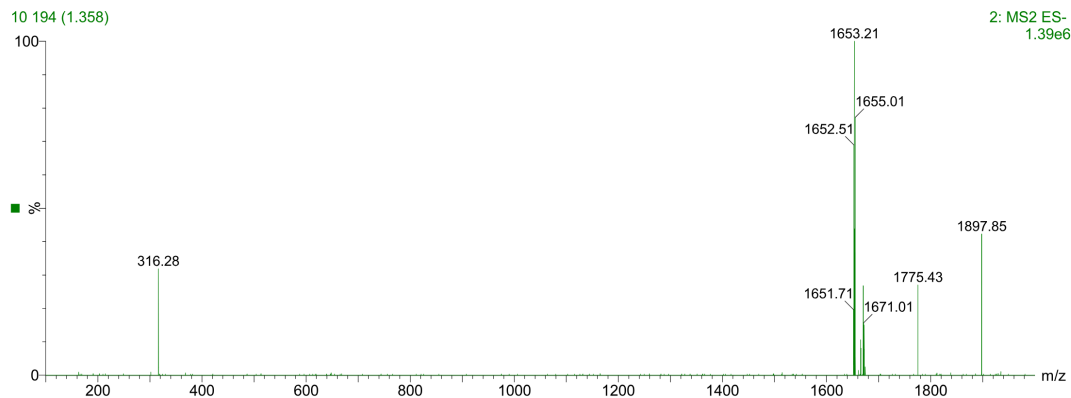
Lot Number: IWR-7964352-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
 Analysis Conducted: 04/16/2026
 Searchable via: horizonanalytical.com

BPC-157 (10mg) • Pubchem CID: 108101
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Pubchem CID: 16132341

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

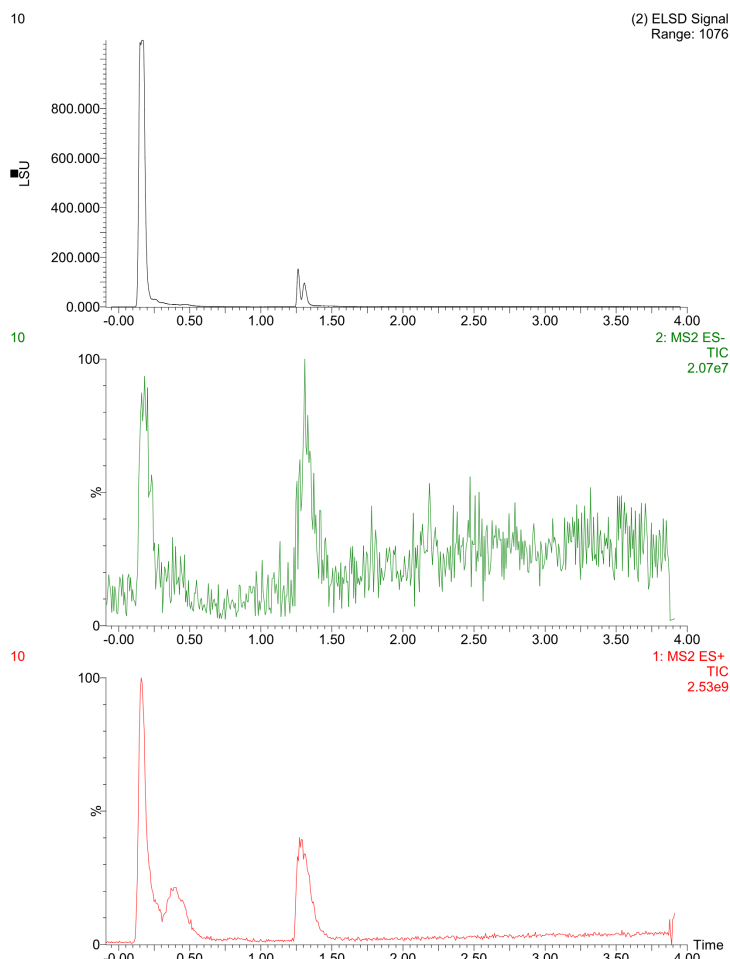


Proudly Owned and Operated in the USA

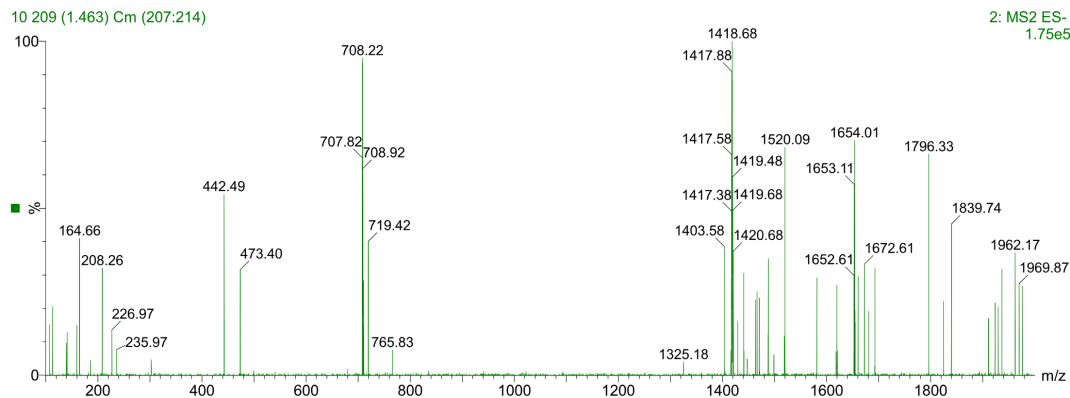
Lot Number: IWR-7964352-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

TB-500 (10mg) • Pubchem CID: 16132341
Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: **IWR-7964352-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **04/21/2026**
 Analysis Conducted: **04/16/2026**
 Searchable via: **horizonanalytical.com**

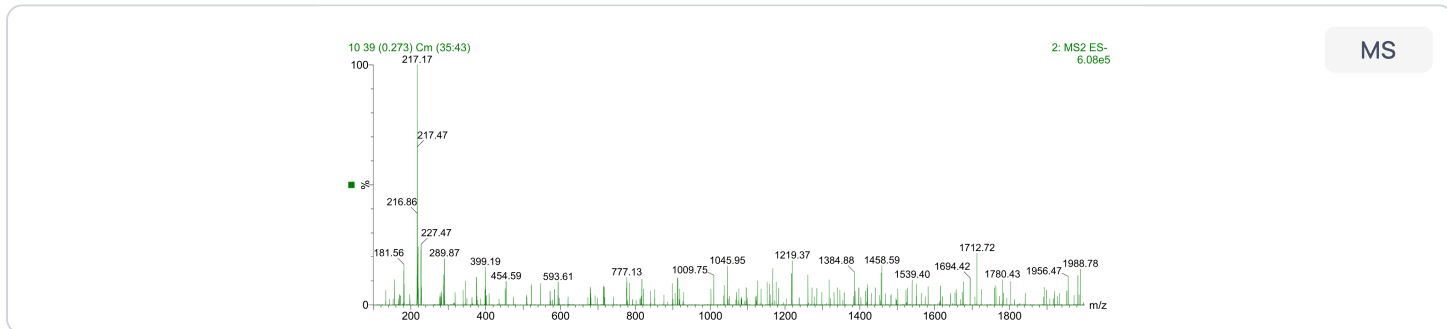
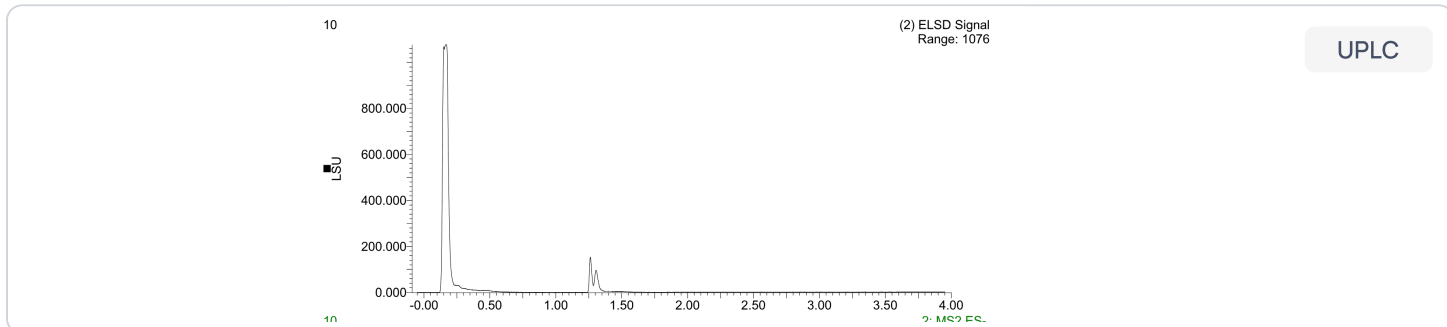
Compound:	GHK-Cu
Lot:	IWR-7964352-P
Appearance:	Blue Lyophilized Powder

CAS:	89030-95-5
Formula:	C ₁₄ H ₂₃ CuN ₆ O ₄
Mol Weight:	~402.92 g/mol

Pubchem CID: 71587328

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	GHK-Cu	GHK-Cu	
Quantity:	50mg	49.1mg	
Purity:	≥ 98%	99.45%	



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

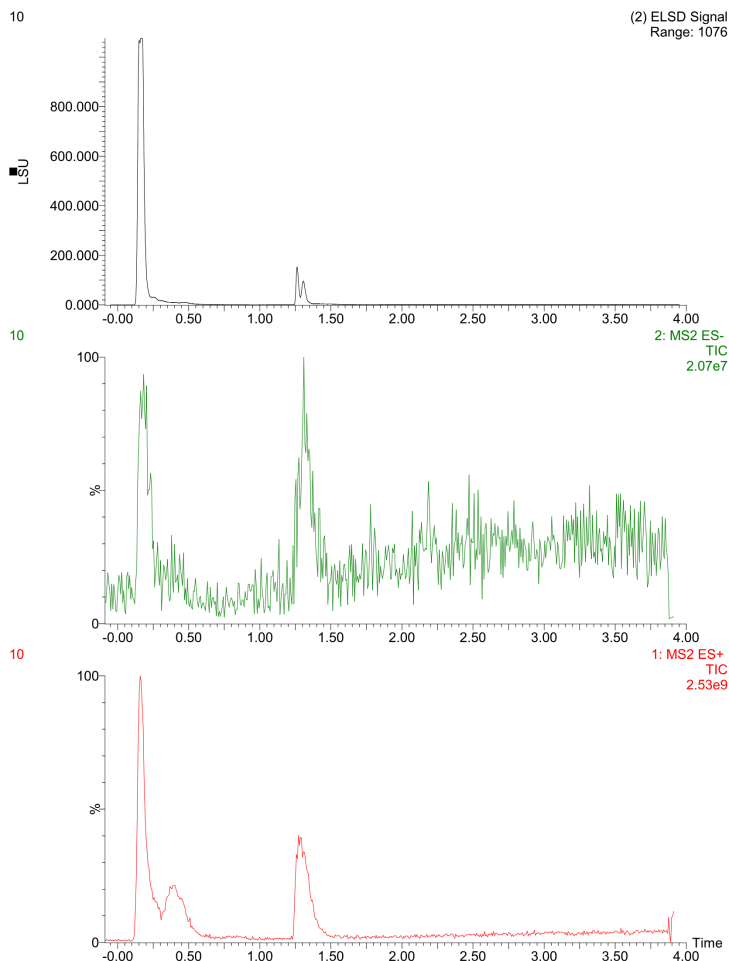


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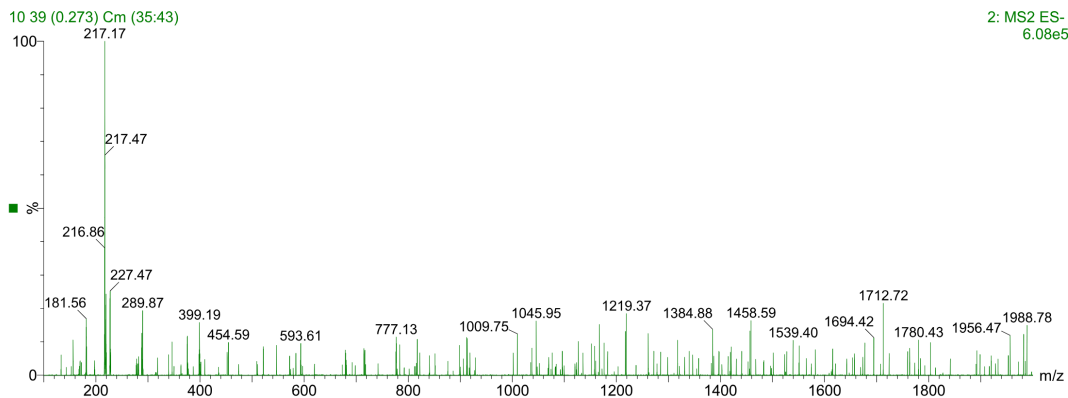
Lot Number: IWR-7964352-P
 Client Name: Iron Within Research
 Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
 Analysis Conducted: 04/16/2026
 Searchable via: horizonanalytical.com

GHK-Cu (50mg) • Pubchem CID: 71587328
 Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Lot Number: **IWR-7964352-P**
 Client Name: **Iron Within Research**
 Identity: **www.ironwithinresearch.com**

Received Date: **04/21/2026**
 Analysis Conducted: **04/16/2026**
 Searchable via: **horizonanalytical.com**

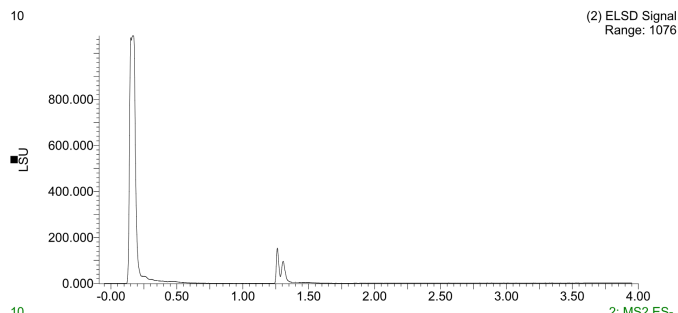
Compound:	KPV
Lot:	IWR-7964352-P
Appearance:	White Lyophilized Powder

CAS:	112965-21-6
Formula:	C ₁₇ H ₃₂ N ₆ O ₄
Mol Weight:	~384.48 g/mol

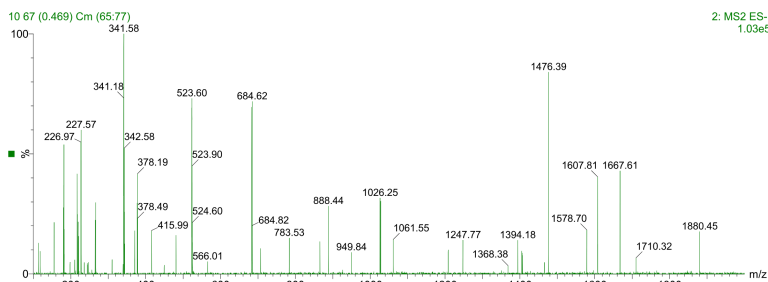
Pubchem CID: 9929972

Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	KPV	KPV	
Quantity:	10mg	10.12mg	
Purity:	≥ 98%	99.05%	



UPLC



MS

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

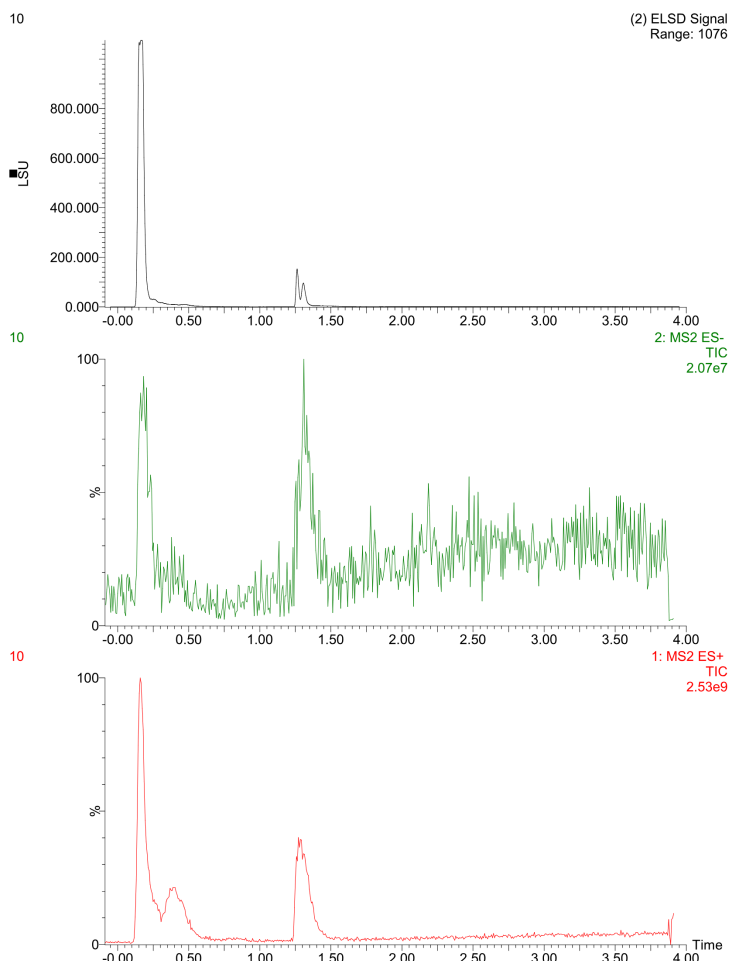


This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Lot Number: IWR-7964352-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

KPV (10mg) • Pubchem CID: 9929972
Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)

